

Switching Basics Theory

Task 1 :- Open Cisco Packet Tracer or any network simulator, create a simple network with one switch and two PCs, and use the 'show mac address-table' command to observe how the MAC address table is initially empty.

ANS :-

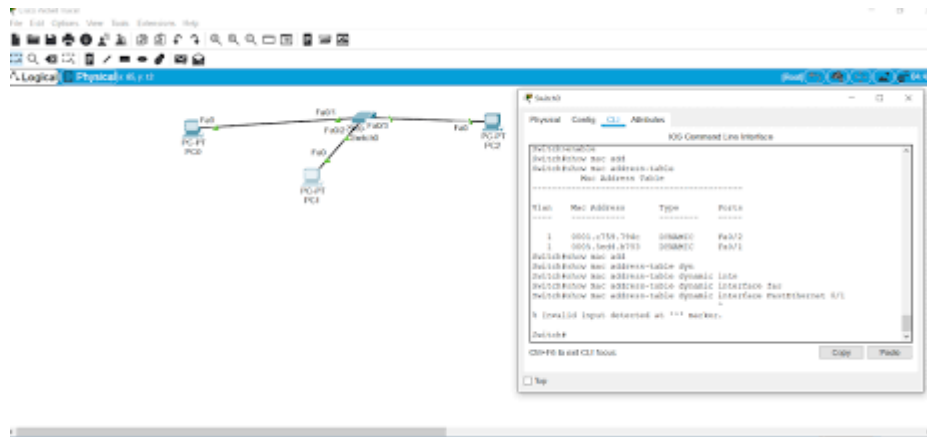
Build the Network : Drag 1 Switch and 2 PCs onto the workspace, then connect both PCs to the switch using copper straight-through cables.

Open Switch CLI : Click the Switch, go to the CLI tab, and press Enter.

Check Table: Type enable, press Enter, then run :

```
show mac address-table
```

Result : The table is completely empty (no dynamic MAC addresses listed) because the PCs haven't sent any traffic yet.



Task 2 :- Send a ping from PC1 to PC2 in your simulated network, then immediately run 'show mac address-table' on the switch and note how the MAC addresses of both PCs appear in the table.

ANS :-

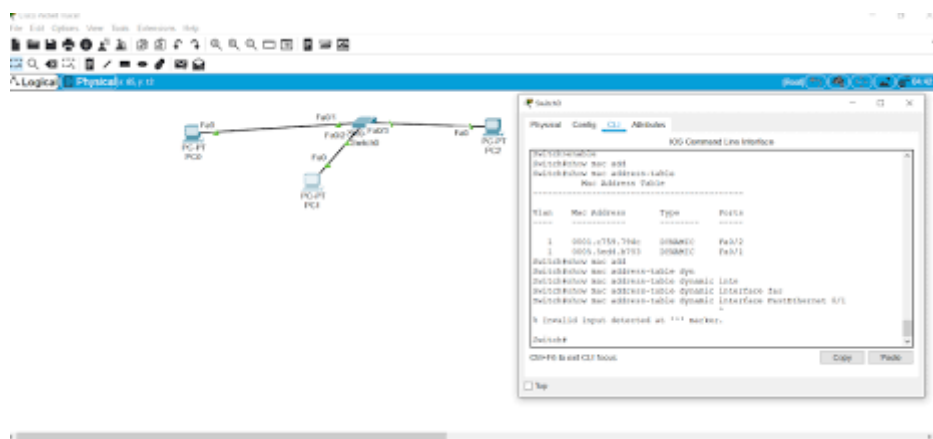
1. **Send Ping :** Click PC1 → Desktop → Command Prompt, then ping PC2 :

```
ping <PC2-IP-Address>
```

2. **Check Switch Table** : Open Switch CLI and run :

```
show mac address-table
```

3. **Result** : Both PCs' MAC addresses and connected ports now appear in the table because the switch learned them from the ping traffic.



Task 3 :- Add a third PC to your network, connect it to the switch, and ping from PC3 to PC1. Observe and record how the MAC address table changes after this communication.

ANS :-

1. **Connect PC3 :** Add PC3 to the workspace and connect it to the switch using a straight-through cable.
2. **Ping PC1 :** Open PC3 → Desktop → Command Prompt and ping PC1:

```
ping <PC1-IP-Address>
```

3. **Check Switch Table:** Open Switch CLI and run:

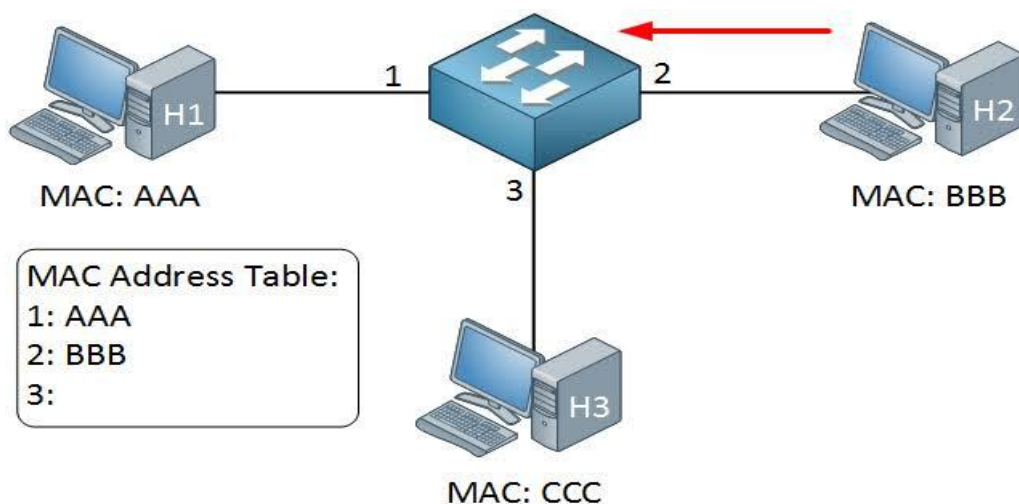
```
show mac address-table
```

4. **Result :** A 3rd MAC address entry is added to the table, linked to PC3's connected switch port. The table now displays all three active PCs.

Task 4 :- Suppose you see the following MAC address table on a switch: Port 1 - 00:11:22:33:44:55, Port 2 - 66:77:88:99:AA:BB. If a frame arrives at the switch on Port 1 with destination MAC 66:77:88:99:AA:BB, explain step-by-step how the switch will forward this frame.

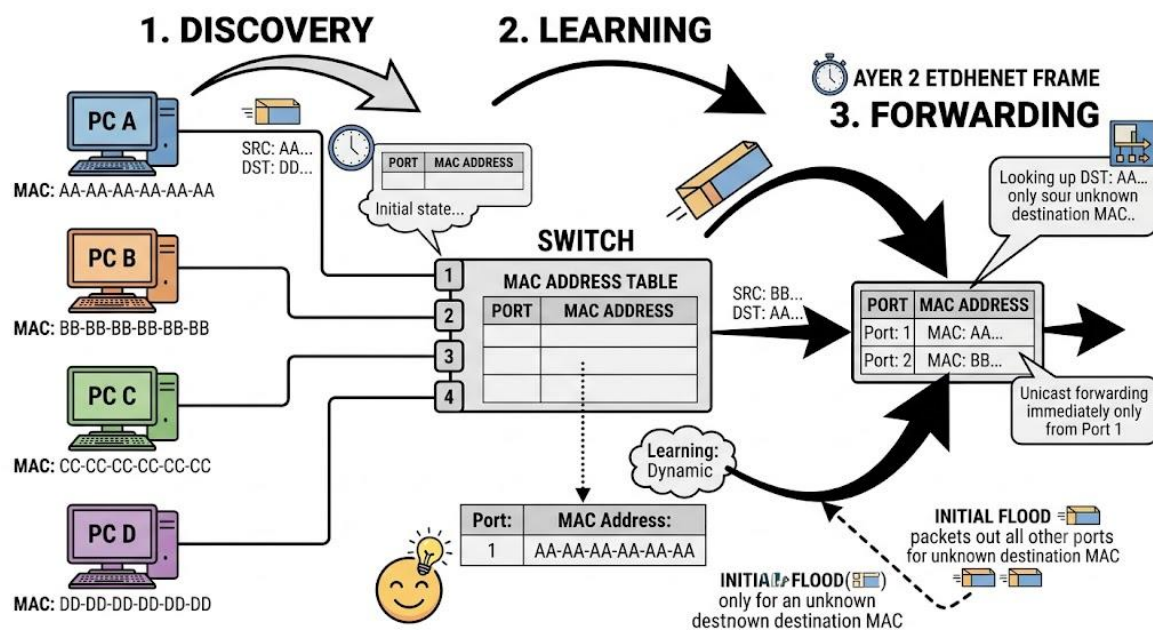
ANS :-

1. **Receives Frame** : The switch gets the frame on Port 1.
2. **Checks Table** : It looks at the destination MAC (66:77:88:99:AA:BB) inside its MAC table.
3. **Finds Match** : It sees 66:77:88:99:AA:BB is mapped directly to Port 2.
4. **Forwards Frame** : The switch sends (unicasts) the frame only out of Port 2 directly to the target device.



Task 5 :- Draw a diagram showing how a switch learns and uses MAC addresses to forward frames in a network with 4 devices, and write a short explanation (3-4 sentences) describing the process in your own words.

ANS :-



When a PC sends data, the switch looks at the sender's MAC address and port number. It adds this info to its memory table, connecting the device to its port. Later, when sending to that device, the switch checks its table and sends it straight to the correct port. If the destination isn't listed, it floods all ports except the sender's to find it.